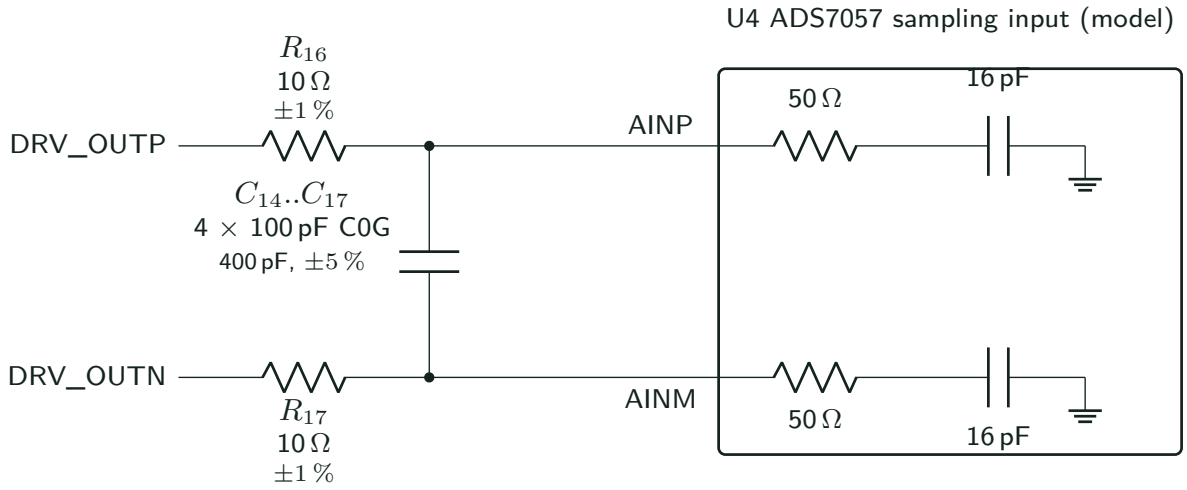
# Slide 8: the ADC input and the ADS7057

A 14-bit ADC and the network that feeds it

ADS7057 test board v3 — study notes

## 1 The circuit




R16 and R17 (10 Ω) and the four 100 pF C0G capacitors C14 to C17 (400 pF) sit between the driver and the ADC. Inside the ADC each input leg is a switch of about 50 Ω in series with a 16 pF sampling capacitor.

## 2 What one code means

Full scale is ±AVDD, and AVDD is also the reference. The code is 14-bit two's complement, so

$$\text{LSB} = \frac{2\,\text{AVDD}}{2^{14}} = \frac{6.6\,\text{V}}{16384} = 0.403\,\text{mV}.$$

A 120 mV event at the head becomes about 0.5 V at the ADC, about 1280 codes, 16 % of the positive range. The sensor noise is 38 to 47 LSB rms, the board adds 0.2 LSB and the ADC adds about 0.6 LSB. The sensor decides the noise floor.

## 3 The kickback

When the sampler closes, the 16 pF capacitor is connected to a node that holds the previous sample. It takes a charge from the input. For the worst step of 0.56 V:

$$Q = C_s\,\Delta V = 16\,\text{pF} \times 0.56\,\text{V} = 9.0\,\text{pC}.$$

The 400 pF bucket supplies this charge, so the input node dips by

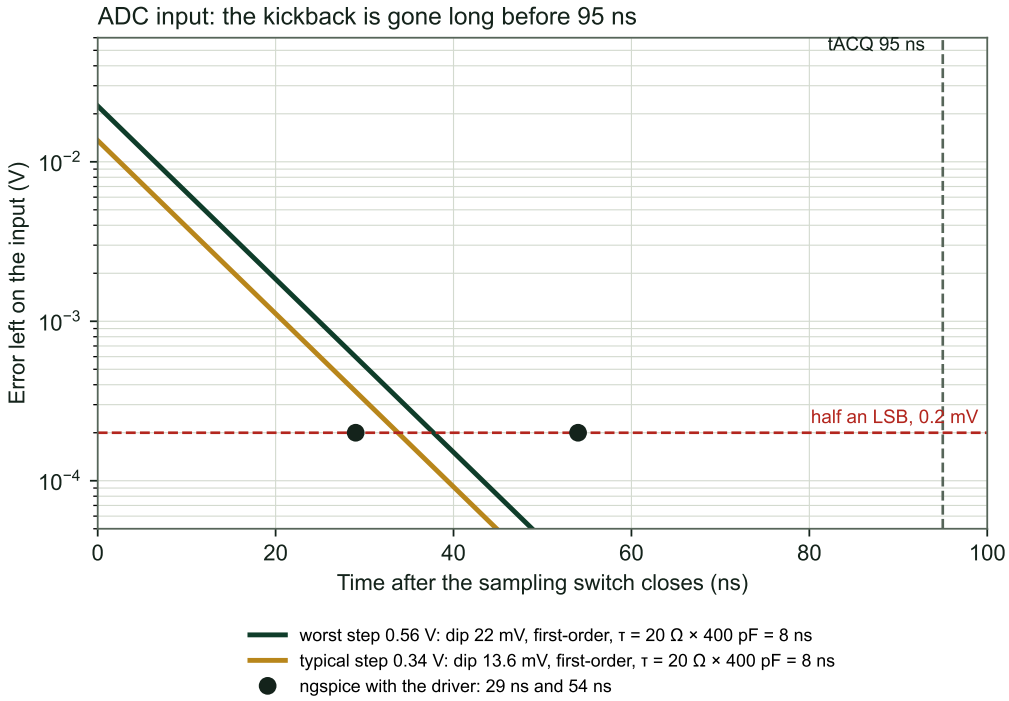
$$\Delta V = \frac{Q}{C_{\text{bucket}}} = \frac{9.0\,\text{pC}}{400\,\text{pF}} = 22.4\,\text{mV}\ (\approx 56\ \text{LSB}).$$

The datasheet asks for at least 20 times the sampling capacitance ($16 \times 20 = 320\,\text{pF}$). The board has 25 times.

## 4 Settling

The two 10 Ω resistors recharge the bucket. First-order model: $\tau = (R_{16} + R_{17})\,C = 20\,\Omega \times 400\,\text{pF} = 8\,\text{ns}$, a pole at $1/(2\pi\tau) = 19.9\,\text{MHz}$, far above the band. The dip decays as $e^{-t/\tau}$. It reaches half an LSB (0.2 mV) after

$$t = \tau \ln \frac{22.4\,\text{mV}}{0.2\,\text{mV}} = 8\,\text{ns} \times 4.7 = 38\,\text{ns}.$$




The ngspice run with the real driver gives 29 ns for a typical step and 54 ns for the worst step. The ADC allows 95 ns. The hand estimate (38 ns) and the simulation are the same size, and both leave margin.

## 5 Decoupling

AVDD has 4.7 µF (C18) and 100 nF (C22). DVDD has 1 µF (C19) and 100 nF (C20). C22 is 2.4 mm and C18 2.7 mm from the ADC. The 100 nF parts for DVDD sit farther (4.5 mm), which is a point to improve.

> The ADS7057 is 14 bit, up to 2.5 MSPS, ±0.9 LSB INL, 79.6 dB SNR at 2 kHz. It samples on the falling edge of CS.